

One-Step Equations

AQA · KS3 Year 8

Answer each question in your workbook. Show your working.

Method

See what was done to the letter Choose the inverse operation Do it to BOTH sides
Simplify to leave the letter alone Check by substituting

Common mistakes

- Only changing one side of the equation.
- Wrong inverse: the inverse of $\times 3$ is $\div 3$, not $- 3$.
- For $x \div 4 = 3$, multiply both sides by 4.
- Sign slips: $x + 7 = 2$ gives $x = -5$.

Worked example

Solve $x + 7 = 12$.

Inverse of $+ 7$ is $- 7$: $x = 12 - 7 = 5$.

Check: $5 + 7 = 12$

Visual aid

Visual Example

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



Balance scale: $x + 7$ on the left pan equals 12 on the right pan. Remove 7 from both pans so the scale stays level, leaving $x = 5$.

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○

Solve: a) $x + 5 = 12$ b) $y - 3 = 10$ c) $m + 8 = 8$

Q2 of 8

[2] ● ○ ○ ○ ○

Solve: a) $4x = 20$ b) $x \div 3 = 6$ c) $7x = 56$

Q3 of 8

[2] ● ● ○ ○ ○

Solve: a) $x - 9 = -2$ b) $x + 11 = 4$ c) $6x = -30$

Q4 of 8

[3] ● ● ○ ○ ○

Solve: a) $x \div 5 = -4$ b) $-3x = 21$ c) $x + 2.5 = 6$

Q5 of 8

[3] ● ● ● ○ ○

Solve (letter on the right): a) $9 = x + 4$ b) $20 = 5x$ c) $-8 = x - 3$

Q6 of 8

[3] ● ● ● ○ ○

Form and solve: a) A number n times 4 is 28.
b) 5 subtracted from p gives 12.

Q7 of 8

[3] ● ● ● ● ○

Rectangle: width w , length $5w = 35$ cm. a) Equation in w . b) Find w . c) Perimeter.

Q8 of 8

[5] ● ● ● ● ●

Each parcel costs $\pounds x$; 4 parcels cost $\pounds 26$. a) Equation. b) Price of one parcel. c) Most whole parcels for $\pounds 40$?